

- (3.) **Critical:** Drop by drop, add an equal volume of 2 × HEPES-buffered saline. Vortex after each drop. Do not let the calcium complex precipitate!
- (4.) Incubate at room temperature for 15 min.
- (5.) Apply dropwise to cells. Rock gently; avoid swirling.
- Critical:** Calcium-mediated uptake depends on temperature and pH. Keep medium pH stable and avoid cell overgrowth.
- (6.) After 6–12 h, remove the medium. Wash cells with PBS and feed fresh medium.



⌚ 15 min



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Troubleshooting

Transfection with Lipofectamine® 2000

In Step 6:

- High cytotoxicity with low transfection efficiency
 - Replace or supplement serum-free medium with full medium after 3–9 h.
 - Reduce Lipofectamine volume by 40%. Excessive cationic lipid damages membranes without improving DNA delivery.
 - Ensure cells are 30–40% confluent at transfection. Over-confluent cells are less permissive to lipoplex uptake.

Recipes

Polyethyleneimine (PEI), pH 7.0, 1 g/L

Amount	Ingredient	Stock	Final
100 mg	Polyethyleneimine (PEI), linear, transfection-grade [26913-06-4]	25 000 g/mol	
To 100 mL	Water, reagent-grade		

Bring the stirred solution to neutral pH. Be patient, this may take hours! Recheck pH after 15 min to ensure it has not drifted. Filter-sterilize through a 0.22 µm polyethersulfone (PES) membrane to remove undissolved particles. Dispense into 500 µL aliquots. Store at –80 °C. Stable for two months at 4 °C. **Note:** Purchase PEI from Polysciences (Cat. no. 23966).

1 g/L Polyethyleneimine (PEI)
pH 7.0

WARNING



Skin sensitization; Eye irritation; Hazardous to the aquatic environment

Date: Sign: R0131

HEPES-buffered saline, pH 7.1, 2 ×

Amount	Ingredient	Stock	Final
25 mL	HEPES, pH 7.5 ⚡ R0024	1 M	50 mM
28 mL	NaCl ⚡ R0046	5 M	280 mM
0.2 g	Na ₂ HPO ₄ · 7 H ₂ O [7782-85-6]	268.07 g/mol	1.5 mM
5.4 mL	Glucose, sterile-filtered ⚡ R0021	20%	12 mM
To 500 mL	Water, reagent-grade		

Filter sterilize. Stable for 1 year at room temperature. Alternatively, store at –20 °C. **Critical:** The pH is extremely critical for a successful transfection; even 0.1 pH units deviation reduces efficiency!

2 × HEPES-buffered saline
50 mM HEPES, 280 mM NaCl, 1.5 mM Na₂HPO₄,
12 mM Glucose, pH 7.1

Date: Sign: R0132

List of references

Change log

2017-05-15	Moritz Pott	Initial calcium chloride protocol; Daniel Summerer lab.
2020-05-20	Yael David	Initial Lipofectamine® protocol; Tom Muir lab.
⋮	⋮	⋮
2023-02-09	Benjamin C. Buchmuller	Added FuGENE® and PEI protocols.
2023-04-04	Benjamin C. Buchmuller	Updated Lipofectamine® volumes.
2025-04-05	Benjamin C. Buchmuller	Harmonized provisions. Updated safety information for PEI.

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